

Order of Magnitude

Answer key · 10 marks total

1. $30 \div 3 = 10$ (1) **one order of magnitude** bigger (1)
2. $(5 \times 10^{-5}) \div (5 \times 10^{-6}) = 10$ (1) **10 times wider** (1)
3. $(2 \times 10^{-6}) \div (2 \times 10^{-8}) = 100$ (1) **two orders of magnitude** bigger (1)
4. $40 \div 4 = 10$ **times bigger** (1) — award both marks for correct division and answer (1)
5. $(1 \times 10^{-5}) \div (1 \times 10^{-7}) = 100$ (1) **two orders of magnitude** (1)

Marking note: "order of magnitude" — a ratio of 10 is one order of magnitude, a ratio of 100 is two orders of magnitude, and so on. Award method marks for correct division even if the order-of-magnitude wording is missing.